

Midterm Exam

CS 101 Algorithmic Problem Solving L2

Fall 2023

Name(s): _____

HU ID (*e.g.*, *xy01042*): _____

Exam Instructions

- **Exam Duration:** 2 hours and 30 minutes
- **Number of Questions:** This exam contains 4 coding questions and 1 debugging question.
- **Technical Requirements:** You are not allowed to use code editors, compilers, or online resources during this exam. Only electronic calculators are permitted during this exam.
- **Communication:** During the exam, do not communicate with anyone except the instructor or exam proctors if you have questions about the exam.
- **Honesty:** Academic honesty is expected. Do not engage in any form of cheating, plagiarism, or unauthorized collaboration.
- **Answer Format:** Write your answers clearly in the provided spaces. Do not use a Pencil.
- **Time Management:** Monitor the time carefully and pace yourself. Do not spend too much time on any one question or section.
- **Exam Completion:** Before submitting, take a moment to review your answers to ensure you have completed all questions and that your responses are clear and accurate.

Good Luck!

1. Last Word

Given a string `s` consisting of words and spaces, return the length of the last word in the string.

A word is a maximal sub string consisting of non-space characters only.

Write a function `Last word()`, that inputs the string with spaces and prints the last word and its length as shown in the sample below.

Constraints

- `isinstance(s,str)`

Interaction

The input consists a string containing spaces and characters and calculates the length of the last word in the string and prints the last word and its length.

Sample

Input	Output
Hello World	The last word is 'World' with length 5.
peek a boo, I see you	The last word is 'you' with length 3.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
This is an easy question	The last word is 'question' with length 8.
what do you say?	The last word is 'say?' with length 4.

Problem Identification

Briefly explain the problem you identified in the above question that led you to your solution.

Pseudocode

```

1 def lastWord():
2     s=input("Enter String:")
3
4     word=""
5     count=0
6     for i in range(len(s)-1,0,-1):
7         if s[i] != " ":
8             count+=1
9             word+=s[i]
10        if s[i]==" ":
11            break
12    print("The Last word is","",word[::-1],"", " with length",count)
13 lastWord()

```

Dry Run

Using any **one** input provided in the **Exercise section** above, dry run your pseudocode in the space below.

2. Faulty Keyboard

Sana decided to make a program to calculate the factorial of a number. But the user has a faulty keyboard, random things are input before the actual integer whose factorial is needed, and the actual integer is always followed by the '@' symbol, Write a function *FixFact* that takes no parameter and finds the number whose factorial is needed. This function then sends this number to another function *fact* that takes an integer *n* and returns the factorial of *n* to the calling function which prints it.

Constraints

-
- $1 \leq n \leq 99$

Interaction

The input comprises a single line containing a single string denoting the value of *n*.

The output contains the product of the odd digits in *n*.

Sample

Input	Output
hjk	120
jkj	
7	
hjk	
5	
@	
ab67	720
jdk	
6	
@	

In the first case, 5 is the integer which is followed by the '@' character so, a factorial of 5 is calculated which is 120.

In the second case, the desired number is 6, so the factorial of 6 is calculated which is 720.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
456	6
df@	
jhd	
3	
@	
3	24
342d	
4	
@	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode

```

1 def fact(j):
2     facto=1
3     if j == 0:
4         return 1
5     else:
6         for i in range(1,j+1):
7             facto=facto*i
8         return facto
9
10 def FixFact():

```

```
11     a = ''
12     j = 0
13     while a != '@':
14         j = a
15         a = input()
16     print(fact(int(j)))
17 FixFact()
```

Dry Run

Using any **one** of the inputs provided in the **Exercise section** above, dry run your pseudocode in the space below.

3. Product of Odds

Write a Python program to compute the product of the odd digits in a given number.

Write a function named `odd_products` that takes a parameter `n` and returns the product of its odd digits. you can treat this number as an integer or as a string.

Constraints

- `isinstance(n, str)`

Interaction

The input comprises a single line containing a single string denoting the value of n .

The output contains the product of the odd digits in n .

Sample

Input	Output
12345	15
2456781	70

In the first case, $n = 12345$. so you need to calculate the product of 1,3 and 5 which is 15

In the second case, $n = 2456781$. so you need to calculate the product of 2,5,7,1 which is 70

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
456	5
7896654	1512

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode

```
1 def odd_products(n):
2     product=1
3     n=str(n)
4     for i in range(0,len(n),2):
5         product=product*int(n[i])
6     return product
7
8 print(odd_products(12345))
```

Dry Run

Using any **one** of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

4. shift Digits

Write a Python program to shift the digits in a number n to the left, wrapping the extra digits around. If the no of shifts is greater than the number of digits in n , reverse the string.

Write a function `shift_digits` that takes 2 parameters n and `shift` where n is the number to be shifted and `shift` is the no of left shifts needed to be done.

Note: you can convert this number to a string if needed.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq \text{shift} \leq 10^5$

Interaction

The input comprises a single line containing two space-separated integers n and `shift` respectively.

The output contains the digit after applying left circular shifts to it.

Sample

Input	Output
12345 3	45123
567843 7	348765

In the first case, the no of left shifts is 3 so the first 3 digits (123) are shifted to the end (after 45) in the same order.

In the second case, the number of shifts is greater than the digits so reverse the number.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2345678 4	6782345
10234570 5	57010234

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode

```
1 def shift_digits(n, shifts):
2     if (shifts > len(n)):
3         print(n[::-1])
4     else:
5         left = n[0:shifts]
6         right = n[shifts:]
```

```
7         print(right+left)
8
9     shift_digits('10234570',5)
```

Dry Run

Using any **one** of the inputs provided in the **Exercise section** above, dry run your pseudocode in the space below.

COMPLETE AND FIX

5. Geometric Series

The Geometric Series is a sequence of numbers, where each term after the first is found by multiplying the previous one by a fixed, non-zero number called the common ratio.

2, 6, 18, 54, 162, 486, 1458, 4374, ...

write a function *Geometric_series* that takes the following parameters and calculates and prints the respective Geometric series.

num - represents the starting value in the geometric series

ratio - represents the common ratio in geometric series

limit - represents the total number of values that need to be printed in the series

Constraints

- $1 \leq num \leq 10$
- $1 \leq ratio \leq 99$
- $1 \leq limit \leq 50$

Interaction

The input comprises a single line containing three space separated integers named num, ratio, and limit. Output contains the geometric series starting from num and contacting limit terms. Note the Output must contains the comma between the terms with the last term ending with a full stop.

Sample

Input	Output
2 3 5	2,6,18,54,162.
3 2 6	3,6,12,24,48,96.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
3 4 4	
5 5 8	

Proposed Solution below is a partial and buggy implementation done by Sarah. You have to Help her fix the Function below to meet the requirements mentioned above and fix errors if any.

```
1 def Geometric_series(num,ratio,limit):
2
3     while(limit>1):
4         num*ratio
5         print(str(num)+',')
6         limit-=1
7
```

Dry Run

Using one of the inputs provided in the Sample section above, dry run the proposed code solution below.

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and the errors in each line.

Correct Solution

Rewrite the function `Geometric_series` with the correct results meeting all the requirements.

```
1
2 def Geometric_series(num,ratio,limit):
3     print(num,end=",")
4     while(limit>1):
5         num=num*ratio
6         if(limit-1==1):
7             print(num,end=".")
8         else:
9             print(num,end=",")
10        limit-=1
11
12 Geometric_series(2,3,5)
13
```

Rough Work